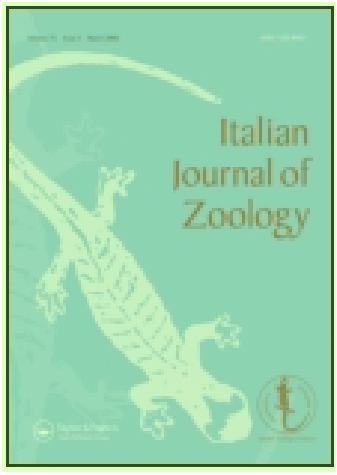
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Fruits and roots: Wild boar foods during the cold season in the southwestern Pyrenees 9

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Fruits and roots: wild boar foods during the cold season in the southwestern Pyrenees

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ABSTRACT

In order to understand wild boar's diet during the cold season and its alternatives to mast in a southern European mountain range, 234 wild boar stomachs we collected in autumn and winter during three consecutive hunting seasons. The study was undertaken in several valleys of the southwestern Pyrenees, a third of which was forest. Results indicated that diet mainly consisted of plant matter from the aerial parts of different forest plants. The most important items in terms of abundance and frequency of appearance were beechnuts *Fagus sylvatica*, acorns from white oak *Quercus humilis* and holm oak *Q. ilex*, and bracken *Pteridium aquilinum* roots. Those items represented 70.7% of the total volume. The alternative to hard mast was roots. The amount of animal matter in the diet was small but consumed frequently and consistently during the three hunting periods. Due to the abundance of forest fruits, alternative natural foods and the mosaic of forest, there was little crop damage in the area.

KEY WORDS: Wild boar - Feeding habits - Stomach content - Southwestern Pyrenees - Autumn and winter.

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INTRODUCTION

The wild boar *Sus scrofa* L. 1758 can be found in all continents, except Antarctica, partly due to its original widespread distribution and then to subsequent translocations imposed by man (d'Huart, 1991). Wild boars are quite abundant but heavily hunted, partly because they are highly prized and in part because they can cause crop damage (Vassant *et al.*, 1992). As a result, biological material is easily obtained to analyse aspects such as reproduction, population structure, or diet (Macchi *et al.*, 1995).

Despite their abundance and the availability of samples, our knowledge of wild boars in different environments can be considered scarce. In Europe, some studies have been carried out on food habits in Mediterranean environments (Dardaillon, 1987; Sjarmidi *et al.*, 1992; Abáigar, 1993; Valet *et al.*, 1994) as well as in east-central Europe (Genov, 1981), in agricultural landscapes (Leranoz, 1983; Genov, 1987), and in marsh ecosystems (Garzón *et al.*, 1983; Herrero *et al.*, 2003). However, little is known about their diet in high mountain environments, and most information is derived from two studies in the Alps (Baubet *et al.*, 1997; Durio *et al.*, 1995). This is particularly relevant because the impact boars have on the biocenosis on which they feed, both above ground and underground, can affect ground-nesting birds (Fernández-Llario & Carranza, 1996), soft and hard mast (Sáenz de Buruaga, 1995), soil invertebrates (Baubet *et al.*, 1997) or roots (Mauget *et al.*, 1984). Consequently these animals can have an important impact on communities (Bratton, 1975) and modify soil cycling (Kotanen, 1995), tree regeneration (Lacki & Lancia, 1986) or vegetation composition (Kotanen, 1995).

Little is known about the food strategy of wild boar in the Pyrenees, despite its importance in mountain ecology. In this study we analysed its diet and food alternatives during autumn and winter based on stomach-content analysis.

MATERIALS AND METHODS

Study area

The study was carried out in several north-south oriented valleys in the southwestern Pyrenees (Aragon, Spain). The total surface area was 78,197 ha, a third of which was forest. The elevation ranged from 800 to 2,450 m a.s.l, though hunting battues only took place in the forests between 800 and 1600 m, since higher altitudes were supra-forest pastures. The topography was rough with large contrasts in vegetation due to slope, height and orientation. Mixed forests predominated even if there were also monospecific ones. The landscape was extensively modified by human activity but rural abandonment has allowed the forest to recover in recent decades. The dominant forest species were, in decreasing order, Scots pine *Pinus sylvestris*, beech *Fagus sylvatica*, white oak *Quercus humilis*, fir *Abies alba*, and holm oak *Q. ilex*, the last of which was scarce. Lower altitudes were occupied by white oak and holm oak. Scots pine was present throughout the altitudinal range. Beech represented the altitudinal limit of the forest in most of the area. Fir was distributed among the Scots pine and beech forests. The average density of wild boars for the whole area was

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3.3/km^2 (SE = 0.6, n = 516; Herrero, 2003), and was determined using battues with hounds during the hunting season (before breeding began). This method can be considered the most accurate to estimate ungulate numbers (Puček *et al.*, 1975). Density did not vary among years (Rs = 0.05, P = 0.37, Spearman rank correlation).

The study area is transitional between the Atlantic and the sub-Mediterranean regions, with great ecological diversity. The average annual precipitation at 1,400 m. a.s.l. is about 1,550 mm.

Animals

Human density is less than 4 inhabitants per km^2, agriculture is of little importance but free-ranging livestock (cattle, sheep, cows, horses and goats) are present in large numbers. Other wild ungulates are Pyrenean chamois *Rupicapra p. pyrenaica* (above 1,600 m), with an estimated census of 900 animals (García-González & Hidalgo, 1989), roe deer *Capreolus capreolus* in all the forest habitats and an average of 1.5 animals per km^2 (SE = 2.6, n = 642; Herrero, 2003), which increased during the study period (Rs = 0.15; P = 0.0001) and red deer *Cervus elaphus*, with well below 1 deer per km^2 in the whole area (Herrero, 2003). The only potential predator is brown bear *Ursus arctos* although it is nearly extinct (Caussimont & Herrero, 1997). Wild boars are heavily exploited and about 500 are killed in the area every hunting season. Poaching is also common. There have been no reports of wild boar predation on sheep or other livestock, but they probably damage the ground nests of different avian species.

Sampling technique

Samples were collected from wild boars after hunting battues from October to February in three consecutive hunting seasons (1990/1991, 1991/1992 and 1992/1993). We analysed 234 stomachs: 53 in the first hunting season, 62 in the second, and 119 in the third. Individuals in the population were normally between 6 and 24 months of age, although older animals were also found. The sex-ratio was 1:1. Detailed information regarding the demographic and reproductive characteristics of this population can be found in Herrero (2003).

All the stomachs were stored in 5% formalin. Of all these samples, 19 stomachs were empty (< 50 cm^3), and another eight had been punctured by bullets during capture. None of those were used to calculate total volume but the punctured stomachs were used to count frequency of occurrence of the different food items. The volume of stomach content and its different items was measured to a precision of 10 cm^3. The contents were washed with water using a 1.0-mm mesh to eliminate gastric juices (T. Abáigar, 1990, Ph.D. thesis, University of Navarre) and unidentifiable particles (Wood & Roark, 1980). After drying the contents, 250 cc were chosen randomly for complete identification. Those samples were then washed in four consecutive sieves of decreasing mesh size (8.0, 4.0, 2.0, and 1.0 mm) to facilitate identification by size.

Plant items were normally identified *de visu* after creating a reference collection, or by using keys in the case of *Gramineae* (Aldezabal & García-González, 1992, García Suárez *et al.*, 1993). Animal matter was also identified *de visu* or by mammal hair keys (Faliú *et al.*, 1980).

The volumes of the different food items from each stomach were calculated using a graduated test tube (± 0.1 cm^3). The percentage volume was calculated as the total volume of each food item in all stomachs divided by the total volume. Volumes below 0.1 cm^3 were considered trace amounts. The frequency of occurrence of all food items was calculated by dividing the number of stomachs with a given item by the total number of stomachs.

Fruiting of the three hard mast species (beechnuts, and holm and white oak acorns) was surveyed in the area and classified as abundant, scarce or absent.

Statistics

The average annual proportion of each item in the diet during the three hunting seasons was analysed using a Kruskal-Wallis non-parametric test.

RESULTS

Plant material was the most important food item in the wild boar diet in all three periods (95.2%; Table I). The aerial parts of plants were consumed more (71.4%) than underground parts (23.8%). Mushrooms were not as important (0.4%). Fruits were the preferred part of plants (62.2%), especially hard mast (58.8%) compared to soft mast (3.4%). In the underground portion, bracken represented almost half of the rooted volumes. Animal matter was scarce (3.8%). Vertebrates and invertebrates occurred in similar small proportions (2.7% vs 1.1%, respectively). Mammals were mainly red deer and badger *Meles meles* carrion.

The frequency of the different items (Table I) was predominantly from plants (100%) compared to animal matter (50.7%). The percentages of aerial parts (95.8%), leaves and stems (61.9%), fruits (78.6%) and underground parts (62.3%) were also high. Snails (23.2%) and larvae of different invertebrates (20.9%) were the most frequent animal matter.

TABLE I - *Autumn and winter percentage volume and frequency of occurrence of food items identified in stomachs of wild boars in the southwestern Pyrenees, Spain* (n = *215, 1990-1993). +, traces.*

Food items	Volume (%)	Frequency (%)
PLANT MATERIAL	95.2	100.0
AERIAL PARTS	71.4	95.8
Leaves and stems	8.8	61.9
gramineae	3.3	34.9
Fruits	62.2	78.6
Hard mast	58.8	67.4
Fagus sylvatica	28.9	33.9
Quercus humilis	27.3	41.9
Quercus ilex	2.5	8.4
Soft mast	3.4	20.9
Crataegus monogyna	0.6	7.9
Juniperus communis	0.5	2.3
Malus sp.	1.6	10.2
Rosa sp.	0.7	7.9
Viscum album	+	0.9
Mushrooms	0.4	6.0
UNDERGROUND PARTS	23.8	62.3
Roots	22.2	59.5
Pteridium aquilinum	12.0	35.8
Bulbs	1.6	6.0
ANIMAL MATTER	3.8	50.7
INVERTEBRATES	1.1	44.2
Insects	+	3.2
Snails	0.8	23.2
Earthworms	+	3.7
Larvae	0.2	20.9
VERTEBRATES	2.7	10.2
Mammals	2.5	5.6
Birds	+	1.9
UNDETERMINED	1.0	19.1

Overall, there were no differences in plant and animal consumption among the three hunting periods. However, the consumption of aerial and underground plant parts did vary. The most important hard mast species (beechnuts and of white oak acorns), varied in terms of the consumption of aerial parts (Table II; Kruskal-Wallis test: H = 2, n = 207).

TABLE II - *Autumn and winter wild boar dietary variation in the southwestern Pyrenees (Spain), 1990-1993. Annual average volumes (%); Kruskall-Wallis test H = 2,* n = *207).*

	1990-1991 (n = 47)	1991-1992 (n = 53)	1992-1993 (n = 109)	P
PLANT MATERIAL	95.8	95.4	95.4	NS
AERIAL PARTS	33.2	66.7	86.7	**
Leaves and stems	18.0	14.5	3.9	NS
Gramineae	4.3	6.3	1.8	NS
Fruits	13.9	50.5	82.8	**
Hard mast	10.8	44.3	78.5	**
Fagus sylvatica	8.4 (s)	25.3 (a)	26.8 (a)	*
Quercus humilis	0.0 (s)	18.9 (s)	47.1 (a)	**
Quercus ilex	2.4 (a)	0.0 (s)	4.6 (a)	NS
Soft mast	3.0	6.2	4.2	NS
UNDERGROUND PARTS	62.6	28.7	8.7	**
Roots	60.0	28.5	7.9	**
Pteridium aquilinum	31.3	19.6	4.4	**
ANIMAL MATTER	2.5	4.2	3.4	NS

*, 0.01 < P < 0.05; **, P < 0.01; (a), abundant fruiting; (s), scarce fruiting.

DISCUSSION

Natural forest fruits were the most important food resource for wild boars in the study area. Our data support the idea that wild boars are highly omnivorous and opportunistic. Plant material comprised most of the diet but inter-annual variations can be observed depending on food availability (Schley & Roper, 2003). Boars prefer natural resources to cultivated plants and crops if there are enough of the former (Mackin, 1970, Andrzejewski & Jezierski, 1978; Genov, 1981; Sjarmidi *et al.*, 1992). Small amounts of animal matter were consumed but quite frequently, as in other populations. Wild boars fed on a few abundant preferred foods that are highly digestible and nutritional. In autumn and winter they were mainly frugivorous, as found in other studies (Henry & Conley 1972; Scott & Pelton, 1981; M. Baettig 1980, M.Sc. thesis, University of Vaud; Sjarmidi *et al.*, 1992; Abáigar, 1993). In our study area, *Fagaceae* fruits were normally quite abundant. Oak and beech fruits have a high carbohydrate content that allows the boars to store fat for the more critical season at the end of the winter.

When some hard mast are lacking, boars tend to feed on the remaining ones or on underground resources. Roots, particularly bracken, were the alternative to hard mast in the first period, averaging 60% of total food volume. The production of beechnuts and white oak acorns, which were the most consumed items, varied widely in the three periods, as did consumption. As the fruiting of these species is highly inconsistent, boars must tended to vary their daily movements and habitat use over the years. In this sense, Singer *et al.* (1981) reported altitudinal seasonal migrations to lower zones in winter in Tennessee, although animals occupied the same seasonal ranges in successive years. In the present study, scavenging was of little importance. The diverse landscape (with Atlantic and sub-Mediterranean influences) and different mast production imply alternative resources in an inconsistent environment. This helps to maintain a large wild boar population that feeds mainly on forest products during autumn and winter.

Current forest management in the area favours the development of Scots pine forests, although they are less interesting from an ecological point of view and less diverse and productive for game species (Jedrzejewska *et al.*, 1994). The high forest species richness of this area should be taken into account in planning forest management from the ecological and economic points of view. In recent years hunting has become an economic resource for local communities.

Rooting behaviour probably modifies the plant population dynamics. This can be especially important when there is a lack of hard mast, as most of the food comes from underground, with bracken being preferred.

In autumn wild boars in many habitats can be considered frugivorous. They mainly consume seeds and disperse them. Though, as Genard & Lascourret (1985) report, the percentage of seeds destroyed during digestion depends on their size. Approximately 75% of the hard mast smaller than 4 mm remained undamaged after passing through the digestive tract.

Wild boars preferred forest fruits in the Atlantic and sub-Mediterranean forests of the south-western Pyrenees. Their feeding habits should also be taken in account in planning forest management, as boars are an important ecological factor as well as an important economic and social resource. Their important rooting activity is probably modifying the ecological forest succession and should be studied in more detail.

Finally, there is a clear trophic and spatial overlap (Berducou *et al.* 1983) in autumn between wild boar and the endangered brown bear population. Both species coincide in the same feeding areas. As wild boars are hunted in battues, some bears could be killed during the hunting season.

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